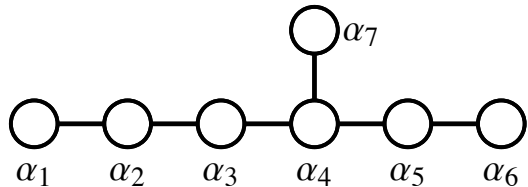


e_7 :

Dynkin Diagram



$$w(\mathbf{ad}) := [0000010]$$

$$w(\mathbf{F}) := [1000000]$$

Cartan Matrix

$$\begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & -1 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 2 \end{pmatrix}$$

Weight Metric

$$\begin{pmatrix} \frac{3}{2} & 2 & \frac{5}{2} & 3 & 2 & 1 & \frac{3}{2} \\ 2 & 4 & 5 & 6 & 4 & 2 & 3 \\ \frac{5}{2} & 5 & \frac{15}{2} & 9 & 6 & 3 & \frac{9}{2} \\ 3 & 6 & 9 & 12 & 8 & 4 & 6 \\ 2 & 4 & 6 & 8 & 6 & 3 & 4 \\ 1 & 2 & 3 & 4 & 3 & 2 & 2 \\ \frac{3}{2} & 3 & \frac{9}{2} & 6 & 4 & 2 & \frac{7}{2} \end{pmatrix}$$